UNIVERSITY OF ILLINOIS URBANA-CHAMPAIGN

STAT 576: Empirical Process Theory and Weak Convergence Fall 2025
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Lecture 10

Content.

1. Hoeffding's Lemma
2. Hoeffding's inequality
3. Chernoff's inequality
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1 Hoeffding's Lemma

Lemma 1 (Hoeffding's Lemma)**.** *Let $X \in [a, b]$ almost surely, with $\mathbb{E}[X] = 0$. Then, for all $s \in \mathbb{R}$,*

$$\mathbb{E}\, e^{sX} \;\leq\; \exp\left(\frac{s^2(b-a)^2}{8}\right).$$

Proof. Define the log-mgf $\psi(s) = \log \mathbb{E}[e^{sX}]$. Note that

- $\psi(0) = 0$
- $\psi'(s) = \frac{\mathbb{E}[Xe^{sX}]}{\mathbb{E}[e^{sX}]}$, $\psi'(0) = \mathbb{E}[X] = 0$
- We now calculate the second moment.

$$\begin{aligned}\psi''(s) &= \frac{\mathbb{E}[X^2e^{sX}]\mathbb{E}[e^{sX}] - (\mathbb{E}[Xe^{sX}])^2}{(\mathbb{E}[e^{sX}])^2} \\ &= \frac{\mathbb{E}[X^2e^{sX}]}{\mathbb{E}[e^{sX}]} - (\frac{\mathbb{E}[Xe^{sX}]}{\mathbb{E}[e^{sX}]})^2\end{aligned}$$

Using change of measure, define a new probability measure $\mathbb{Q}$ such that $d\mathbb{Q} = \frac{e^{sX}}{\mathbb{E}[e^{sX}]} d\mathbb{P}_X$, then we have

$$\mathbb{E}_{X\sim\mathbb{Q}}[f(X)] = \mathbb{E}_{X\sim\mathbb{P}_X}[f(X)\frac{e^{sX}}{\mathbb{E}[e^{sX}]}].^1$$

So, we can see that

$$\begin{aligned}\psi''(s) &= \mathbb{E}_{X\sim\mathbb{Q}}[X^2] - (\mathbb{E}_{X\sim\mathbb{Q}}[X])^2 \\ &= \text{Var}_{\mathbb{Q}}(X) \\ &\leq \frac{(b-a)^2}{4}\end{aligned}$$

where the last step is because $X \in [a, b]$ a.s. under $\mathbb{P}_X$ implies that $X \in [a, b]$ a.s. under $\mathbb{Q}$.

Now, we have $\psi''(s) \leq \frac{(b-a)^2}{4}$, $\psi(0) = 0$, $\psi'(0) = 0$. By the fundamental theorem of calculus, we have

$$\begin{aligned}\psi'(s) - \psi'(0) &= \int_0^s \psi''(t)\, dt \leq \frac{s(b-a)^2}{4} \\ \psi(s) - \psi(0) &= \int_0^s \psi'(t)\, dt \leq \int_0^s \frac{t(b-a)^2}{4}\, dt = \frac{s^2(b-a)^2}{8}\end{aligned}$$

So, $\psi(s) = \log \mathbb{E}[e^{sX}] \leq \frac{s^2(b-a)^2}{8}$, and

$$\mathbb{E}[e^{sX}] \leq \exp(\frac{s^2(b-a)^2}{8})$$

□

2 Hoeffding's inequality

Theorem 1. *Let $X_1, \ldots, X_N$ be independent random variables such that $X_i \in [a_i, b_i]$ a.s., $\forall i$. Then, for any $t \geq 0$:*

$$\mathbb{P}\left[\sum_{i=1}^{N}(X_i - \mathbb{E}[X_i]) \geq t\right] \leq \exp\left(-\frac{2t^2}{\sum_{i=1}^{N}(b_i - a_i)^2}\right)$$

[1] The object $\frac{d\mathbb{Q}}{d\mathbb{P}_X}$ is called the Radon-Nikodym derivative. For any set U, $\mathbb{Q}[U] = \mathbb{P}[\mathbb{1}\{U \in X]e^{sX}]/\mathbb{E}[e^{sX}]$.

Remark 1. *We have*

- *We typically have* $\sum_{i=1}^{N}(b_i - a_i)^2 = O(N)$.
- *Therefore,* $\sum_{i=1}^{N}(X_i - \mathbb{E}[X_i]) = O(\sqrt{N})$.

Proof of Hoeffding's inequality. WLOG, assume $\mathbb{E}[X_i] = 0, \forall i$.[2]

$$
\begin{aligned}
&\mathbb{P}[\sum_{i=1}^{N} X_i \geq t] \\
=&\mathbb{P}[\lambda \sum_{i=1}^{N} X_i \geq \lambda t] && \forall \lambda \geq 0 \\
=&\mathbb{P}[\exp(\lambda \sum_{i=1}^{N} X_i) \geq \exp(\lambda t)] \\
\leq&\mathbb{E}[\exp(\lambda \sum_{i=1}^{N} X_i)]e^{-\lambda t} && \text{by Markov's inequality} \\
\leq&e^{-\lambda t} \prod_{i=1}^{N} \mathbb{E}[\exp(\lambda X_i)] && \text{since } X_i\text{'s are independent} \\
\leq&e^{-\lambda t} \prod_{i=1}^{N} \exp(\frac{\lambda^2 (b_i - a_i)^2}{8}) && \text{by Hoeffding's Lemma} \\
=&\exp(-\lambda t + \frac{\lambda^2}{8} \sum_{i=1}^{N} (b_i - a_i)^2) \\
:=&\exp(f(\lambda))
\end{aligned}
$$

$\lambda^* = \frac{4t}{\sum_{i=1}^{N}(b_i - a_i)^2}$ minimizes $f(\lambda)$, and $f(\lambda^*) = -\frac{2t^2}{\sum_{i=1}^{N}(b_i - a_i)^2}$.[3]

Since the above bound holds for any $\lambda > 0$, we have

$$\mathbb{P}[\sum_{i=1}^{N} X_i \geq t] \leq \exp(f(\lambda^*)) = \exp(-\frac{2t^2}{\sum_{i=1}^{N}(b_i - a_i)^2})$$

□

Remark. *The argument above is known as Cramér-Chernoff Method.*

[2] Note that if we consider instead $X_i' = X_i - \mathbb{E}[X_i]$, then one can establish instead the same claim for X_i' for which the range is $[a_i - \mathbb{E}[X_i], b_i - \mathbb{E}[X_i]]$ that have the same 'length' as $[a_i, b_i]$.

[3] Here, it is crucial that $\lambda^* \geq 0$, as all the argument above pertains to this regime.

Comparing two Tail Bounds

$$\begin{aligned}
&\mathbb{P}[|X| \geq t] = \mathbb{P}[|X|^q \geq t^q] \leq t^{-q}\mathbb{E}[|X|^q] \\
\Rightarrow &\mathbb{P}[|X| \geq t] \leq \inf_{q>0} \; t^{-q}\mathbb{E}[|X|^q] \qquad (1) \\
&\mathbb{P}[|X| \geq t] \leq e^{-\lambda t}\mathbb{E}[e^{\lambda X}] \qquad (1) \\
&\mathbb{P}[|X| \geq t] \leq \inf_{\lambda>0} e^{-\lambda t}\mathbb{E}[e^{\lambda X}] \qquad (2) \qquad (2)
\end{aligned}$$

FACT: Bound (1) is actually better than bound (2).

Remark 2. *Two-sided bounds are also possible*

$$\mathbb{P}[|\sum_{i=1}^{N} X_i - \mathbb{E}[X_i] \geq t|] \leq 2\exp(-\frac{t^2}{\sum_{i=1}^{N}(b_i - a_i)^2})$$

Outline:

$$\mathbb{P}[|X| \geq t] = \mathbb{P}[X \geq t] + \mathbb{P}[X \leq -t]$$

The first term can be controlled through Cramer-Chernoff method, and the second term can be bounded similarly:

$$\begin{aligned}
\mathbb{P}[X \leq -t] &= \mathbb{P}[\lambda X \geq -\lambda t] \qquad\qquad \lambda < 0 \\
&\leq e^{\lambda t}\mathbb{E}[e^{\lambda X}]
\end{aligned}$$

Example : [Failure of Hoeffding's Inequality] Let $X_1, \ldots, X_N \overset{i.i.d.}{\sim}$ Bernoulli($\frac{\lambda}{N}$), then $S_N = \sum_{i=1}^{N} X_i \sim$ Binomial(N, $\frac{\lambda}{N}$), which is known to converge to $\mathrm{Pois}(\lambda)$. This suggests S_N is roughly $O(1)$ for large N.

However, by Hoeffding,

$$\mathbb{P}[|S_N - \lambda| \geq t] \leq 2\exp(-\frac{2t^2}{N})$$

Hoeffding predicts $|S_N - \lambda| = O(\sqrt{N})$. So, $S_N = O(\sqrt{N})$. Hoeffding only cares about the RANGE! The fix is Chernoff's inequality, introduced next.

3 Chernoff's inequality

Theorem 2. *Let X_i be independent Bernoulli(p_i), and*

$$S_N = \sum_{i=1}^{N} X_i \text{ with } \mu = \mathbb{E}[S_N] = \sum_{i=1}^{N} \mathbb{E}[X_i]$$

Then, for any $t > \mu$:

$$\mathbb{P}[S_N \geq t] \leq e^{-\mu}(\frac{e\mu}{t})^t$$

Proof.

$$\begin{aligned}
\mathbb{P}[S_N \geq t] &= \mathbb{P}[\lambda S_N \geq \lambda t] && \forall \lambda > 0 \\
&= \mathbb{P}[e^{\lambda S_n} \geq e^{\lambda t}] \\
&\leq e^{-\lambda t}\mathbb{E}[e^{\lambda S_N}] && \text{Markov' inequality}
\end{aligned}$$

We have $\mathbb{E}[e^{\lambda S_N}] = \prod_{i=1}^{N} \mathbb{E}[e^{\lambda X_i}]$, and

$$\begin{aligned}
\mathbb{E}[e^{\lambda X_i}] &= p_i e^{\lambda} + (1 - p_i) \\
&= 1 + p_i(e^{\lambda} - 1) \\
&\leq e^{p_i(e^{\lambda}-1)}
\end{aligned}$$

where the last step is using the fact that $e^x \geq x + 1$.[4] Combining, we get

$$\begin{aligned}
\mathbb{P}[S_N \geq t] &\leq e^{-\lambda t} \prod_{i=1}^{N} e^{pi(e^{\lambda}-1)} \\
&= \exp(-\lambda t + \sum_{i=1}^{N} p_i(e^{\lambda} - 1)) \\
&= \exp(-\lambda t + \mu(e^{\lambda} - 1)) \\
&:= \exp(f(\lambda))
\end{aligned}$$

Let $f'(\lambda) = 0$, we get that $\lambda^* = \log(\frac{t}{\mu})$ minimizes $f(\lambda)$. Also, we have $\lambda^* > 0$ because $t > \mu$. Plugging λ^* into the previous bound estabilishes the conclusion. □

[4] Note that the inequality gap - namely $e^x - x - 1$ - is small if x is close to zero. As such, the error in this step is minimal when p_i is small, which is in line with the example above.

4 Sub-Gaussian Random Variables

Definition 1. *A random variable X is said to be sub-Gaussian with variance proxy σ^2 if $\mathbb{E}[X] = 0$ and*

$$\mathbb{E}[e^{sX}] \leq \exp(\frac{\sigma^2 s^2}{.} 2)$$

Note that $\exp(\sigma^2 s^2/2)$ is nothing but the MGF of $\mathcal{N}(0, \sigma^2)$.

Lemma 2. *Sub-Gaussian random variables have sub-Gaussian tails.*

Proof.

$$\begin{aligned}
\mathbb{P}[|X| \geq t] &\sim 2\mathbb{P}[X \geq t] \\
\mathbb{P}[X \geq t] &= \mathbb{P}[\lambda X \geq \lambda t] \\
&= \mathbb{P}[e^{\lambda X} \geq e^{\lambda t}] \\
&\leq e^{-\lambda t}\mathbb{E}[e^{\lambda X}] \\
&\leq \exp(-\lambda t + \frac{\sigma^2\lambda^2}{2})
\end{aligned}$$

Choose $\lambda^* = \frac{t}{\sigma^2}$, we have

$$\mathbb{P}[X \geq t] \leq \underbrace{exp(-\frac{t^2}{2\sigma^2})}_{\text{Behaves like } \mathcal{N}(0,\sigma^2)\text{'s tail}}$$

□

Remark 3. *MGF bound $\overset{\text{Cramer-Chernoff}}{\Rightarrow}$ Tail bound $\overset{\text{integral}}{\Rightarrow}$ Moments bound $\overset{\text{Taylor's}}{\Rightarrow}$ MGF bound*

where "integral" refers to

$$\mathbb{E}[|X|] = \int_0^{+\infty} \mathbb{P}[|X| \geq t]\, dt$$

Theorem 3. *Let X_i's be independent Sub-Gaussian random variables with variance proxy σ_i^2. Then, for any $c_1, \ldots, c_N$:*

$$T := \sum_{i=1}^{N} c_i X_i \sim \textit{sub-Gaussian}(\sum_{i=1}^{N} \sigma_i^2 c_i^2)$$

Proof.

$$\begin{aligned}
\mathbb{E}[e^{sT}] &= \mathbb{E}[e^{s\sum_{i=1}^{N} c_i X_i}] \\
&= \prod_i \mathbb{E}[e^{sc_i X_i}] \\
&\leq \exp(\frac{1}{2} s^2 \sum_{i=1}^{N} c_i^2 \sigma_i^2)
\end{aligned}$$

□

Corollary 1. *In the setting of the previous theorem:*

$$\mathbb{P}[|\sum_{i=1}^{N} c_i X_i| \geq t] \leq 2\exp(-\frac{t^2}{2\sum_{i=1}^{N} c_i^2 \sigma_i^2})$$

Recalling that the probability law of two independent random variables is nothing but the convolution of the individual probability laws, we deduce that the class of sub-Gaussian distributions is closed under convolution.